

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester VI Examination, 2022-23

Paper No. CHB-603: Organic Chemistry-V

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer five questions in total, including Question 1 which is compulsory.***Question 1.** Answer all parts of the following: [5 × 2 = 10]

- (a) Differentiate between thermoplastic and thermosetting polymers with suitable examples.
- (b) Discuss the stability of α -helix and β -sheet structures of proteins.
- (c) Write the products of methylation followed by hydrolysis with dilute HCl of D-glucopyranose.
- (d) Show the structure of Thyroxine.
- (e) Why do amino acids have high melting points and are soluble in water? Explain.

Question 2.

- (a) Write down the products of the following amino acid reactions: [6]
 - (i) Alanine $\xrightarrow{\text{LiAlH}_4}$?
 - (ii) Alanine $\xrightarrow{\text{Ph-NCO, then HCl}}$?
- (b) Discuss the cationic and anionic polymerization mechanisms of vinyl monomers. [8]

Question 3.

- (a) Provide a suitable mechanism for the following benzoin condensation reaction: [6]
$$2\text{Benzaldehyde} \xrightarrow{\text{KCN}} \text{Benzoin}$$
- (b) Elucidate the structure of Citral-A using chemical reactions. How will you synthesize Citral starting from methylheptenone? [8]

Question 4. Write short notes on the following: [3 × 4.5 = 13.5]

- (a) Fischer Indole synthesis and its mechanism
- (b) Electrophilic substitution reactions of quinoline and isoquinoline
- (c) Structure and functions of DNA

Question 5.

- (a) Establish the structure of thyroxine with the help of chemical reactions. [6]
- (b) How do you prove the zwitterionic nature of amino acids? Discuss the significance of the isoelectric point. [8]